

BUSINESS INTELLIGENCE AND DATA WAREHOUSING TRIAL QUESTIONS

SET TWO.

1. A database table contains attributes that depend on multiple independent multi-valued facts about the same entity. Which normal form is being violated?
 - a. 2NF
 - b. 3NF
 - c. 4NF
 - d. BCNF
2. In normalization theory, a lossless decomposition ensures that:
 - a. Data is duplicated
 - b. No data is lost after splitting tables
 - c. Tables cannot be joined again
 - d. Redundancy increases
3. Which situation best describes a functional dependency in a relational database?
 - a. One attribute determines another attribute
 - b. Two tables are merged
 - c. Data is encrypted
 - d. Rows are duplicated
4. A table is in BCNF when:
 - a. It has no redundancy
 - b. Every determinant is a candidate key
 - c. It has no primary key
 - d. It is unnormalized
5. In dimensional modeling, surrogate keys are mainly used to:
 - a. Increase redundancy
 - b. Uniquely identify dimension records
 - c. Replace fact tables
 - d. Store measures
6. A slowly changing dimension (SCD) problem occurs when:
 - a. Data never changes
 - b. Dimension attributes change over time
 - c. Fact tables are deleted
 - d. Data is encrypted

7. In a star schema, performance is generally faster because:
 - a. Data is fully normalized
 - b. Fewer joins are required
 - c. Tables are encrypted
 - d. Data is duplicated
8. Snowflake schema increases complexity mainly because:
 - a. It removes fact tables
 - b. It normalizes dimension tables into multiple levels
 - c. It removes relationships
 - d. It deletes data
9. ETL transformation stage primarily ensures:
 - a. Raw data is stored
 - b. Data is cleaned and standardized
 - c. Data is deleted
 - d. Data is archived only
10. Incremental ETL loading is preferred because it:
 - a. Reloads all data every time
 - b. Processes only new or changed data
 - c. Deletes old data
 - d. Avoids transformation
11. Data inconsistency in warehouses is most likely caused by:
 - a. Normalization
 - b. Poor ETL processes
 - c. Indexing
 - d. Aggregation
12. BI systems are most effective when data is:
 - a. Unstructured only
 - b. Integrated and cleaned
 - c. Duplicated
 - d. Random
13. OLAP systems differ from OLTP mainly because OLAP focuses on:
 - a. Transactions
 - b. Analytical queries
 - c. Data entry
 - d. File storage

14. Drill-down in BI allows users to:
- a. Aggregate data
 - b. View detailed levels of data
 - c. Delete data
 - d. Encrypt data
15. KPIs are used in BI mainly to:
- a. Store raw data
 - b. Measure organizational performance
 - c. Design databases
 - d. Normalize tables
16. A fact table without dimension keys will lead to:
- a. Better performance
 - b. Orphan records and poor analysis
 - c. Faster queries
 - d. No impact
17. A multivalued dependency exists when:
- a. One attribute depends on another
 - b. One attribute determines multiple independent attributes
 - c. Data is encrypted
 - d. Tables are merged
18. Which of the following best describes denormalization?
- a. Breaking tables into smaller ones
 - b. Combining tables to improve performance
 - c. Removing keys
 - d. Encrypting data
19. Data mining differs from BI because it focuses on:
- a. Reporting only
 - b. Discovering hidden patterns
 - c. Storing data
 - d. Formatting tables
20. A key risk in poorly designed ETL pipelines is:
- a. Improved performance
 - b. Data loss and inconsistency
 - c. Faster queries
 - d. Reduced storage

SECTION B: STRUCTURED QUESTIONS (40 MARKS)

QUESTION ONE (10 MARKS)

a) A company stores customer orders in a single table containing customer details, product details and order details repeatedly.

Identify two anomalies present in this design. (4 marks)

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b) Explain how decomposition into 3NF tables would solve the problem above. (4 marks)

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c) State two advantages of achieving BCNF over 3NF. (2 marks)

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QUESTION TWO (10 MARKS)

a) A retail company wants to analyze sales across time, region, and product categories. Designing a dimensional model, explain what would be a fact table and three-dimension tables. (4 marks)

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b) Differentiate between star schema and snowflake schema in terms of performance and complexity. (4 marks)

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(ii) A company extracts data from Excel files, APIs and databases into a warehouse.
Explain why ETL is necessary in such an environment. (7 marks)

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b) Describe how errors such as duplicates and missing values are handled in the transform stage. (4 marks)

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c) State and explain the major challenges in large-scale ETL systems. (6 marks)

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QUESTION FOUR (10 MARKS)

a) A hospital wants to analyze patient admission trends and treatment success rates. Explain how Business Intelligence supports decision-making in this scenario. (3 marks)

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b) Differentiate between OLTP and OLAP systems using real-life examples. (4 marks)

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c) Explain how dashboards, KPIs and data visualization work together in BI systems. (3 marks)

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END ////////////////////////////////// TRIAL EXAM ////////// I ////////// WISH ////////////////////////////////// SUCCESS.